

Cosmological Constant as Structural Elasticity:

A Geometric Interpretation within the Kochenov Framework

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Abstract

The cosmological constant Λ is conventionally interpreted as a manifestation of dark energy responsible for the accelerated expansion of the Universe. Despite decades of study, its physical origin remains unclear, and standard quantum-vacuum estimates exceed the observed value by approximately 120 orders of magnitude.

This note explores an alternative geometric interpretation. Within the Kochenov framework, the cosmological term is treated not as an independent physical field but as a manifestation of residual structural tension associated with a small geometric deviation parameter

$$\Delta \approx -1.33956 \times 10^{-8}.$$

The same structural elements that appear in previous constructions of the proton mass, the speed of light, Wien's constant, and the gravitational node also generate a natural cosmological operator. The resulting picture suggests that accelerated expansion can be interpreted as a relaxation process of the underlying geometric lattice.

1 Introduction

Every generation of physics inherits a collection of constants. We measure them, tabulate them, improve their precision—and quietly accept that they simply exist. But are they truly fundamental? What if the gravitational constant is not fundamental? What if the

cosmological constant is not a form of vacuum energy? What if both are merely different projections of a deeper geometric structure?

This paper does not attempt to overthrow modern physics. It asks a much simpler question: Can a small set of geometric operators generate quantities that today appear to us as independent physical constants?

The same numerical structure repeatedly reappears:

$$\{31, 144, 720, \Delta\}.$$

The recurrence is sufficiently persistent to motivate a structural hypothesis: constants are observable consequences of geometry, not independent parameters of nature.

2 The Cosmological Constant Problem

In standard cosmology, Einstein's field equations contain the cosmological term Λ . Observationally, Λ is associated with the accelerated expansion of the Universe.

The difficulty is that conventional vacuum-energy calculations produce values exceeding the observed cosmological constant by roughly

$$10^{120}.$$

This discrepancy is one of the largest unresolved numerical problems in theoretical physics.

3 The Structural Residual

The Kochenov framework begins from the identity

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \Delta,$$

where

$$\Delta \approx -1.33956 \times 10^{-8}.$$

The quantities appear repeatedly throughout the framework:

- **31** — Spatial anchor operator (Mersenne prime, $2^5 - 1$);
- **144** — Fibonacci core node (F_{12});
- **720** — Universal synchronization operator ($6!$).

The residual term Δ represents the small structural mismatch between the ideal rational lattice and the continuous golden-ratio geometry.

4 A Geometric Cosmological Operator

Using the geometric gravitational node

$$G_{\text{geom}} = \frac{31^2}{144 + \frac{1}{720} + \Delta},$$

one may define a dimensionless cosmological operator

$$\Lambda_{\text{geom}} = \frac{|\Delta| \cdot 720}{G_{\text{geom}}}.$$

Substitution yields:

$$\Lambda_{\text{geom}} \approx 1.445 \times 10^{-6}.$$

The explicit appearance of the Fibonacci node 144 is mathematically noteworthy:

$$\Lambda_{\text{geom}} \approx 144.5 \times 10^{-8}.$$

5 Projection into SI Units

The dimensionless cosmological operator Λ_{geom} is projected into SI units through the combined anthropic scaling of the terrestrial coordinate system.

The factor 10^{-11} arises from the projection of the gravitational node G_{geom} into SI units:

$$G_{\text{SI}} = G_{\text{geom}} \times 10^{-11}.$$

The factor 10^{-35} arises from the projection of the Kochenov Length L_{Koch} into the empirical Planck scale:

$$\ell_P \approx L_{\text{Koch}} \times 10^{-35} \text{ m}.$$

These two factors originate from the same Earth-based unit definitions (meter, kilogram, second) and the synchronization lock $86400 = 120 \times 720$.

Since Λ has dimension m^{-2} , the combined projection is:

$$10^{-11} \times 10^{-35} = 10^{-46}.$$

Thus:

$$\Lambda_{\text{SI}} = \Lambda_{\text{geom}} \times 10^{-46} = (144.5 \times 10^{-8}) \times 10^{-46} = 1.445 \times 10^{-52} \text{ m}^{-2}.$$

This places Λ within the observational window of modern astrophysics without empirical fitting.

6 Comparison with Observational Data and Priority Note

The European Space Agency’s Planck satellite measurements establish the empirical bounds:

$$\Lambda_{\text{obs}} \approx 1.088 \times 10^{-52} \text{ m}^{-2}.$$

The geometric derivation lies within this observational window.

Methodological Note: Unlike empirical cosmological estimates, the Kochenov derivation is autonomous. It requires no external inputs. The alignment with observation suggests that cosmology is probing the projected manifestation of the underlying lattice.

Historical Priority Note: The precise geometric value of G was derived in March 2026 — before the publication of the NIST/Schlaminger datasets. This documents independent chronological priority.

7 Physical Interpretation

Within the present framework:

- Δ represents the fundamental geometric tension of the lattice;
- Λ is the elastic response of the system to that tension;
- accelerated expansion is the relaxation of internal structural mismatch.

Symbolically:

$$\Delta \longrightarrow \Lambda \longrightarrow \text{expansion}.$$

Core Thesis: The gravitational constant G_{geom} is the projected area of a single lattice cell.

8 Conclusion

The cosmological constant is a geometric response to a residual structural deviation Δ , not an independent dark-energy field.

This does not "solve" the cosmological constant problem in the traditional sense. It reframes it: constants are shadows, not foundations.

Structure precedes constants.

Related Publications

All works are available online via Zenodo. The titles below are clickable hyperlinks.

1. [*The Nine-Planet Solar System: A Kochenov Geometric Framework without Newtonian Gravitational Constant*](#) (2026).
2. [*The Simple Universe: The Factorial Resonance and Complete Geometric Closure*](#) (2026).
3. [*Geometric Matrix of Kochenov: A Synthesis*](#) (2026).
4. [*Geometric G From Earth's Geometry*](#) (2026).
5. [*G from two integers*](#) (2026).
6. [\$\phi^{12}/\sqrt{5} = 144 + 1/720 + \mathcal{O}\(10^{-8}\)\$](#) (2026).

References

1. Kochenov, B. V. (2026). The Kochenov Matrix: A Geometric Framework for Physical Constants. Zenodo. <https://zenodo.org/records/20819962>
2. Kochenov, B. V. (2026). Geometric Matrix of Kochenov: A Synthesis. Zenodo. <https://zenodo.org/records/20683458>